MAE 3240 Heat Transfer

Spring 2026

Homework Cover Page

Name: Qing Yi Chen Date of submission: 3/3/2026 Problem set number: #4

Due date: 3/3/2026

Acknowledgement:

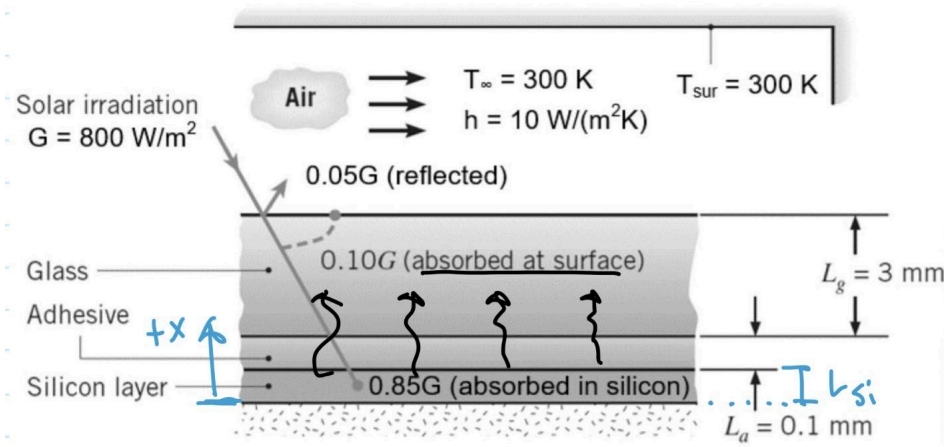
Lecture/course notes and textbook.

Question 1 Homework Solution

Known: Lg = 3mm = 0.003m, kg = 1.5 W/(m*K), La = 0.1mm = 0.0001m, ka = 1 W/(m*K), silicon layer very thin, R''c = 10^-4 m^2 *K/W, no contact resistance between glass and adhesive, backside of silicon layer insulated, T_inf = 300K, h = 10 W/(m^2 * K), Tsurr = 300K, h_rad = 5 W/(m^2 * K), G = 800 W/m^2 (5% reflected from top surface, 10% absorbed top surface, 85% transmitted and absorbed in silicon layer), η = a - bTsi where a = 0.553 and b = 0.001 K^-1, solar energy absorbed in silicon turned to thermal energy and electricity

Find: (a) Show the thermal resistance network that can describe heat transfer across the PV panel. Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network. (b) Calculate the temperatures of the top surface of the glass Tg,top and the silicon layer TSi. (c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

Schematic:



Properties:

Given in known and written down in part (a)

Assumptions: Written down in part (a) too

Analysis:

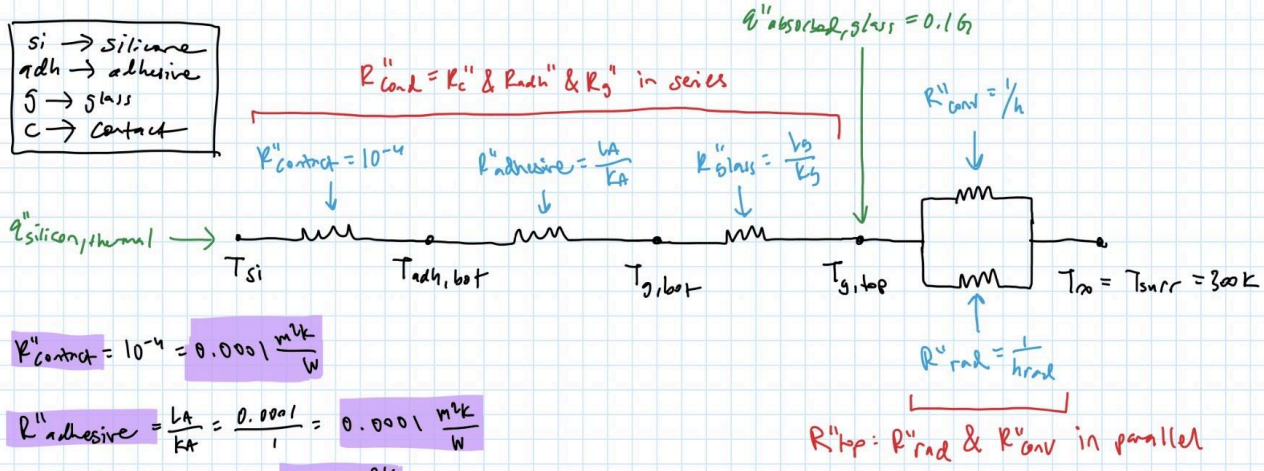
(a) <u>Key assumptions:</u>

- 1D steady state heat transfer (heat flows uniformly up the layers)
- Constant properties
- Adiabatic back surface (problem says back of silicon layer is thermally insulated)

- Isothermal silicon layer (problem says the layer super thin → assume uniform temperature Tsi and negligible internal thermal resistance)
- Assume convection and radiation in environment happen in parallel that can be turned into an equivalent single heat transfer because T_inf = Tsurr = 300K

(a)

si → silicone
adh → adhesive
g → glass
c → contact



$R''_{contact} = 10^{-4} = 0.0001 \frac{m^2K}{W}$

$R''_{adhesive} = \frac{L_A}{k_A} = \frac{0.0001}{1} = 0.0001 \frac{m^2K}{W}$

$R''_{glass} = L_g/k_g = \frac{0.003}{1.5} = 0.002 \frac{m^2K}{W}$

$R''_{conv} = 1/h = 1/10 = 0.1 \frac{m^2K}{W}$

$R''_{rad} = 1/h_{rad} = 1/5 = 0.2 \frac{m^2K}{W}$

$R''_{top} = \left(\frac{1}{R''_{conv}} + \frac{1}{R''_{rad}}\right)^{-1} = (10+5)^{-1} = \frac{1}{15} = 0.06667 \frac{m^2K}{W}$

$R''_{cond} = R_c'' + R_{adh}'' + R_g'' = 0.0001 + 0.0001 + 0.002 = 0.0022 \frac{m^2K}{W}$

$R''_{total} = R''_{cond} + R''_{top} = 0.0667 + 0.0022 = 0.06887 \frac{m^2K}{W}$

(b)

(b) <u>energy balance on panel:</u>

$G = 800\ W/m^2 \longrightarrow q''_{abs,g} = 0.1G = 0.1 \cdot 800 = 80\ W/m^2$

$q''_{si,tot} = 0.85G = 0.85 \cdot 800 = 680\ W/m^2$

$P_{electric} = \eta(0.85G)$

$q''_{si,th} = 0.85G(1-\eta) = 680(1-\eta)$

$q''_{rejected} = q''_{si,th} + q''_{abs,g} = 680(1-\eta) + 80$

$q''_{rej} = \dfrac{T_{s,top} - T_\infty}{R''_{top}} \longrightarrow T_{s,top} = T_\infty + (q''_{rej} \cdot R''_{top})$

$T_{g,top} = 300 + \frac{1}{15}\left[680(1-\eta) + 80\right]$

<u>energy balance between silicon & glass top:</u>

$q''_{si,th} = \dfrac{T_{si} - T_{g,top}}{R''_c + R''_{adh} + R''_{glass}}$

$680(1-\eta) = \dfrac{T_{si} - T_{g,top}}{0.0001 + 0.0001 + 0.002} = \dfrac{T_{si} - T_{g,top}}{0.0022}$

$T_{si} - T_{g,top} = 0.0022\left[680(1-\eta)\right] = 1.496(1-\eta)$

<u>put together & solve:</u>

$\eta = 0.553 - 0.001T_{si}$

$1-\eta = 1 - (0.553 - 0.001T_{si}) = 0.447 + 0.001T_{si}$

$T_{g,top} = T_{si} - 1.496(0.447 + 0.001T_{si})$

$= T_{si} - 0.668712 - 0.0014967T_{si} = -0.668712 + 0.998504T_{si}$

$T_{g,top} = 300 + 1/15\left[680(0.447 + 0.001T_{si}) + 80\right]$

$= 300 + 1/15\left[303.96 + 0.68T_{si} + 80\right] = 300 + 1/15\left[383.96 + 0.68T_{si}\right]$

$= 300 + 25.6 + 0.0453T_{si} = 325.6 + 0.0453T_{si}$

$T_{g,top} = -0.668712 + 0.998504T_{si} = 325.6 + 0.0453T_{si}$

$\longrightarrow 0.953T_{si} = 326.268 \longrightarrow T_{si} = 342.3K$

$T_{g,top} = 325.6 + 0.0453(342.3) = 341.11473K$

$T_{g,top} = 341.11K$ $\quad$ $T_{si} = 342.3K$

(c)

(c) $\eta = 0.553 - 0.001T_{si} = 0.553 - 0.001(342.3) = 0.2107$

$\eta = 0.2107$ or 21.07%

$P''_{electrical} = \eta(0.85G) = 0.2107 \cdot 0.85 \cdot 800 = 143.3\ W/m^2$

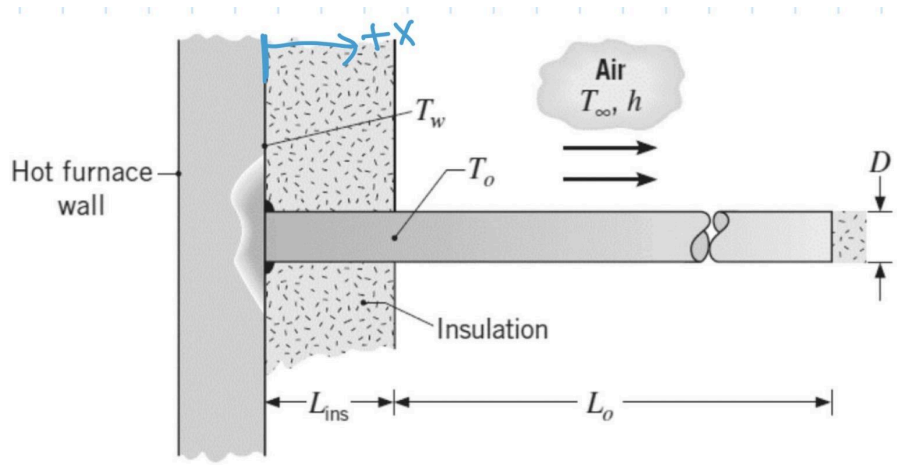
$P''_{electrical} = 143.3\ W/m^2$

Question 2 Homework Solution

Known: D = 10mm = 0.01m, k = 80 W/(m*K), Tw = 200C, Lins = 200mm = 0.2m, Lo, Tmax = 100C, T_inf = 20C, h = 20 W/(m^2 * K)

Find: (a) Derive an expression for the exposed surface temperature To as a function of the prescribed thermal geometrical parameters. (b) If exposed length Lo is 200 mm, calculate the corresponding exposed surface temperature To. Does the exposed length meet the specified operating limit? (c) Calculate the minimum exposed length Lo that ensures To below the specified operating limit.

Schematic:



Properties: given in known section

Assumptions: (1) 1D steady state heat conduction (2) constant properties (3) section of the rod inside the insulation is perfectly insulated on its sides (4) exposed section acts as a fin with an adiabatic tip

Analysis:

(a)

(a) T_o : temp @ exposed part ($x = L_{ins}$) $\quad A_c = \frac{\pi D^2}{4} \quad P = \pi D$

$$q_{cond} = \frac{T_w - T_o}{R_{ins}} = \frac{T_w - T_o}{\frac{L_{ins}}{kA_c}} = \frac{kA_c(T_w - T_o)}{L_{ins}} \quad \text{insulated}$$

$$q_{fin} = \frac{T_o - T_\infty}{R_{fin}} = \frac{T_o - T_\infty}{\frac{1}{\tanh(mL_o)\sqrt{hPkA_c}}} = \left[\tanh(mL_o)\sqrt{hPkA_c}\right](T_o - T_\infty) \quad \text{exposed}$$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{4h}{kD}}$$

$$q_{cond} = q_{fin} \longrightarrow \frac{kA_c}{L_{ins}}(T_w - T_o) = \sqrt{hPkA_c}(T_o - T_\infty)\tanh(mL_o)$$

$$T_w - T_o = L_{ins}\left(\frac{\sqrt{hPkA_c}}{kA_c}\right)\tanh(mL_o)(T_o - T_\infty)$$

$$\frac{\sqrt{hPkA_c}}{kA_c} \longrightarrow \sqrt{\frac{hP}{kA_c}} = m \longrightarrow T_w - T_o = mL_{ins}\tanh(mL_o)(T_o - T_\infty)$$

$$T_w - T_o = mL_{ins}\tanh(mL_o)T_o - mL_{ins}\tanh(mL_o)T_\infty$$

$$T_w + mL_{ins}\tanh(mL_o)T_\infty = T_o\left[1 + mL_{ins}\tanh(mL_o)\right]$$

$$T_o = \frac{T_w + mL_{ins}\tanh(mL_o)T_\infty}{1 + mL_{ins}\tanh(mL_o)} \longrightarrow m = \sqrt{\frac{4h}{kD}}$$

$$T_o = \frac{T_w + L_{ins}T_\infty\tanh\left(L_o\sqrt{\frac{4h}{kD}}\right)\cdot\sqrt{\frac{4h}{kD}}}{1 + L_{ins}\tanh\left(L_o\sqrt{\frac{4h}{kD}}\right)\cdot\sqrt{\frac{4h}{kD}}}$$

Parameters given:

$D, k, T_w, L_{ins}, L_o, T_\infty, h$ ✓

(b)

(b) $L_o = 200\ mm$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{4h}{kD}} = \sqrt{\frac{4 \cdot 20}{50 \cdot 0.01}} = \sqrt{100} = 10\ m^{-1}$$

$$mL_o = 10 \cdot 0.2 = 2 \longrightarrow \tanh(mL_o) = \tanh(2) = 0.964$$

$$mL_{ins} = 10 \cdot 0.2 = 2$$

$$T_o = \frac{T_w + mL_{ins}\tanh(mL_o)T_\infty}{1 + mL_{ins}\tanh(mL_o)} = \frac{200 + 2(0.964)\cdot 20}{1 + 2(0.964)} = 81.47$$

$T_o = 81.47°C \longrightarrow$ Yes meets specified op. operating limit b/c $81.47°C < 100°C$

(c)

(c) $T_0 = 100°C \longrightarrow 100 = \dfrac{200 + 2\tanh(10L_0)\cdot 20}{1 + 2\tanh(10L_0)}$

$X = \tanh(10L_0)$ to make easier $\longrightarrow 100 = \dfrac{200 + 40X}{1 + 2X}$

$100 + 200X = 200 + 40X \longrightarrow 160X = 100 \longrightarrow X = 10/16 = 5/8 = 0.625$

$0.625 = \tanh(10L_0) \longrightarrow \operatorname{arctanh}(0.625) = 10L_0 \longrightarrow \dfrac{\tanh^{-1}(0.625)}{10} = L_0$

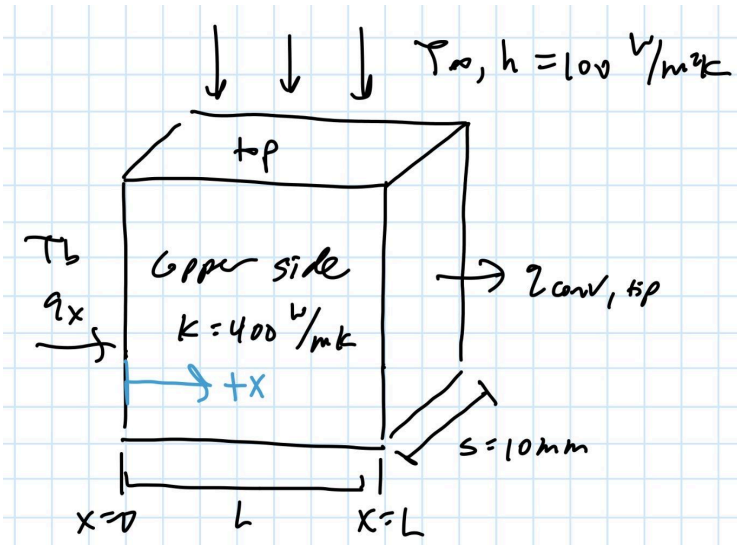
$L_0 = 0.0733m \longrightarrow$ $L_0 = 73.32\ mm$

Question 3 Homework Solution

Known: square pin fin with uniform cross section, s = 10mm = 0.01m, k = 400 W/(m*K), h = 100 W/(m^2 * K), ηf = 0.6, convection tip

Find: (a) Calculate the fin length L. (b) Calculate the thermal resistance of the fin Rf and the fin effectiveness εf.

Schematic:



Properties: all above in the known section

Assumptions: (1) steady state 1D conduction (2) constant properties (3) uniform convection coefficient h over the entire surface

Analysis:

(a)

(a) $A_c = s^2 = (0.01)^2 = 1\times10^{-4}\,m^2$

$P = 4s = 4(0.01) = 0.04\,m$

$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{100\cdot 0.04}{400\cdot 1\times10^{-4}}} = \sqrt{\frac{4}{0.04}} = \sqrt{100} = 10\,m^{-1}$

$L_c = L + A_c/P = L + \frac{1\times10^{-4}}{0.04} = L + 0.0025\,m$

$\eta_f = \frac{\tanh(mL_c)}{mL_c} = 0.6 \rightarrow X = mL_c$ to make easier

$0.6 = \frac{\tanh(X)}{X} \rightarrow 0.6X = \tanh(X) \rightarrow$ use desmos calculator/graphing

$X = mL_c = 1.51222$

$L_c = \frac{1.51222}{m} = \frac{1.51222}{10} = 0.151222\,m$

$L_c = L + 0.0025\,m = 0.151222\,m \rightarrow L = 0.151222 - 0.0025 = 0.148722\,m$

$L = 148.72\,mm$

(b)

(b) $A_{fin} = PL_c = 0.04\cdot 0.151222 = 0.00605\,m^2$

$R_f = \frac{1}{\eta_f h A_{fin}} = \frac{1}{0.6\cdot 100\cdot 0.00605} = 2.755\,K/W$

$R_f = 2.755\,K/W$

$\varepsilon_f = \eta_f\left(\frac{A_{fin}}{A_c}\right) = 0.6\left(\frac{0.00605}{1\times10^{-4}}\right) = 36.29$

or $\varepsilon_f = \frac{R_{base}}{R_f} = \frac{1/hA_c}{R_f} = \frac{1}{hA_cR_f} = \frac{1}{100\cdot 1\times10^{-4}\cdot 2.755} = 36.29$ ✓

$\varepsilon_f = 36.29$